
Algorithm 1 Provable and Notarizable First-in First-out Broadcast with sender p_s : PNFIFO-BC, code for process p_i who runs the protocol in consecutive slot number k with predicate Q .

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Initialize:  $k \leftarrow 1$ 
 $flag_s \leftarrow 0$ 
 $Log_s \leftarrow \{\}$ 

1: function PNFIFO-BCs[ $k$ ]( $v_k$ ) 第k个input: 本地第k个tx
2:   multicast PROPOSAL( $s, k, v_k$ )

3: upon receiving PROPOSAL( $s, k, v_k$ ) from  $p_s$  for the first time do
4:   wait  $flag_s = 0$  and  $Q(v_k) = 1$  do
5:      $\sigma_{k,i} \leftarrow \text{TS.SigShare}_{2f+1}(s, k, v_k)$ 
6:     send VOTE( $s, k, \sigma_{k,i}$ ) to the sender  $p_s$ 
7:      $flag_s \leftarrow 1$ 

8: upon receiving VOTE( $s, k, \sigma_{k,j}$ ) from  $p_j$  for the first time do
9:   if TS.VrfShare $_{2f+1}(s, k, v_k, \sigma_{k,j}) = 1$  do
10:     $T_k \leftarrow T_k \cup (j, \sigma_{k,j})$ 
11:    if  $|T_k| = 2f + 1$  do
12:       $\sigma_k \leftarrow \text{TS.Comb}_{2f+1}(s, k, v_k, T_k)$ 
13:      multicast FINAL( $s, k, \sigma_k$ )

14: upon receiving FINAL( $s, k, \sigma_k$ ) from  $p_s$  for the first time do
15:   wait  $flag_s = 1$  do
16:     if TS.Vfy $_{2f+1}(s, k, v_k, \sigma_k) = 1$  do
17:        $Log_s[k] \leftarrow (v_k, \sigma_k)$ 
18:        $flag_s \leftarrow 0$  and  $k \leftarrow k + 1$ 

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$\triangleright$ slot number
 $\triangleright$ lock flag
 $\triangleright$ output
 $\triangleright$ for sender
 $\triangleright$ for all processes
 $\triangleright$ sign with p_i 's threshold sk
 $\triangleright$ send vote
 $\triangleright$ lock other slots
 $\triangleright$ for sender
 $\triangleright$ verify vote msg
 $\triangleright$ collect vote
 $\triangleright 2f + 1$ valid votes
 $\triangleright$ generate a proof
 $\triangleright$ for all processes
 $\triangleright$ wait until v_k is received
 $\triangleright$ verify message
 $\triangleright$ store k^{th} output
 $\triangleright$ into next slot